

Artificial Intelligence in Pediatric Radiology

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CHLA Radiology Grand Rounds

October 8, 2026

Objectives

- Describe the big players in radiology AI, both general and pediatric
- Summarize what radiology AI does well and what remains unresolved

Methods

Journals, Preprints: PubMed, Embase, arXiv, medRxiv

Conferences: NeurIPS, ICLR, ICML, CVPR, MICCAI, MIDL

Datasets: Hand-curated list

Newsletters: The Imaging Wire, RSNA News, TLDR AI, TLDR Tech, Signify Research, ESR / ECR, Radiology Business

Commercial: FDA list of AI-enabled devices

Date range: January 1 2008 – September 30 2026

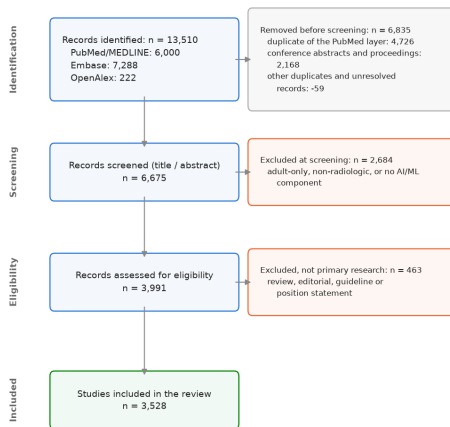
Search strategy: radiology AND AI [AND pediatric]

radiology: (radiology[tiab] OR radiological[tiab] OR radiograph*[tiab] OR "medical imaging"[tiab] OR "diagnostic imaging"[tiab] OR tomography[tiab] OR "magnetic resonance"[tiab] OR MRI[tiab] OR "MR imaging"[tiab] OR "MR images"[tiab] OR MRIs[tiab] OR "MR image"[tiab] OR "MR scan"[tiab] OR "MR scans"[tiab] OR "computed tomography"[tiab] OR CT[tiab] OR ultrasound[tiab] OR ultrasonograph*[tiab] OR sonograph*[tiab] OR echocardiograph*[tiab] OR mammograph*[tiab] OR "chest x-ray"[tiab] OR "x-ray"[tiab] OR fluoroscop*[tiab] OR scintigraph*[tiab] OR "positron emission"[tiab] OR PET[tiab] OR SPECT[tiab] OR angiograph*[tiab] OR neuroimaging[tiab] OR tractograph*[tiab] OR elastograph*[tiab] OR "fMRI"[tiab] OR "functional MRI"[tiab] OR "functional magnetic resonance"[tiab] OR connectome*[tiab] OR "diffusion tensor"[tiab] OR DTI[tiab] OR "bone age"[tiab] OR "skeletal age"[tiab] OR "skeletal maturity"[tiab] OR "barium enema"[tiab])

AI: ("artificial intelligence"[tiab] OR "machine learning"[tiab] OR "deep learning"[tiab] OR "convolutional neural network"[tiab] OR "convolutional neural networks"[tiab] OR "neural network"[tiab] OR "neural networks"[tiab] OR "computer-aided diagnosis"[tiab] OR "computer aided diagnosis"[tiab] OR "computer-aided detection"[tiab] OR radiomic*[tiab] OR "computer vision"[tiab] OR "large language model"[tiab] OR "large language models"[tiab] OR "foundation model"[tiab] OR "foundation models"[tiab] OR "transfer learning"[tiab] OR "vision transformer"[tiab] OR "self-supervised"[tiab] OR "random forest"[tiab] OR "support vector machine"[tiab] OR "gradient boosting"[tiab] OR "natural language processing"[tiab])

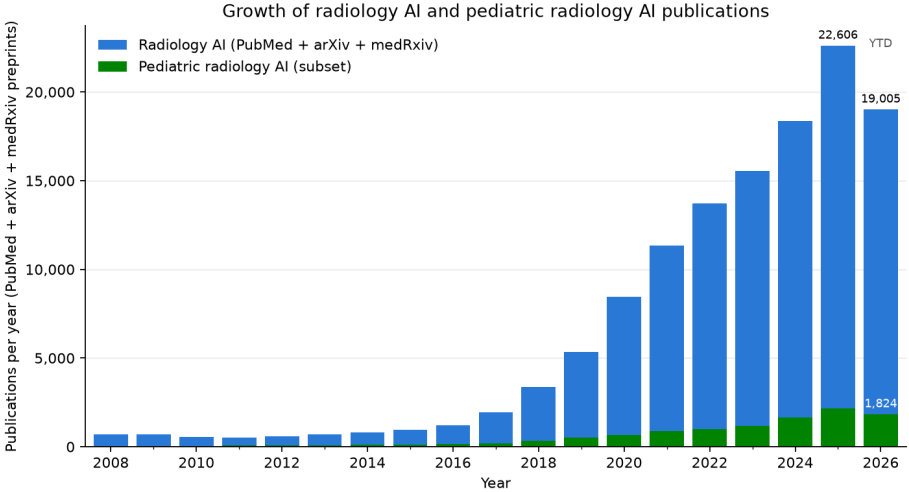
pediatric: (pediatric*[tiab] OR paediatric*[tiab] OR child*[tiab] OR infant*[tiab] OR neonat*[tiab] OR adolescen*[tiab] OR fetal[tiab] OR foetal[tiab] OR fetus*[tiab] OR foetus*[tiab] OR newborn*[tiab] OR preterm[tiab] OR "premature infant"[tiab] OR "premature infants"[tiab] OR "premature birth"[tiab] OR "premature neonate"[tiab] OR "premature neonates"[tiab] OR prenatal[tiab] OR antenatal[tiab] OR perinatal[tiab] OR youth[tiab] OR juvenile[tiab] OR "children's hospital"[tiab] OR schoolchild*[tiab] OR toddler*[tiab])

Study selection

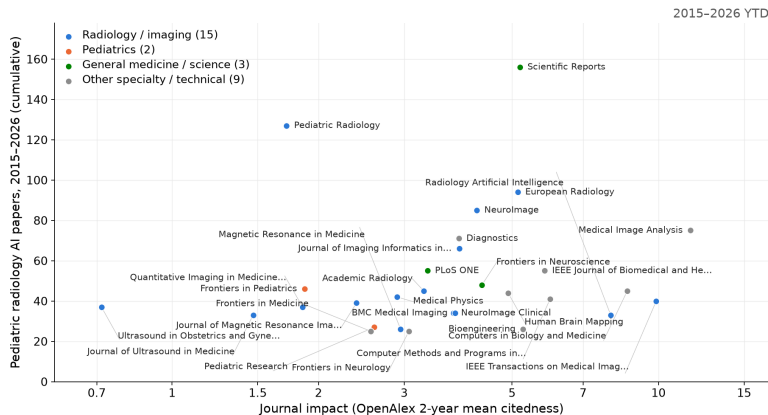


PubMed and Embase layers complete (2005-2026). Conference material is retrieved and reported but excluded by the publication-form criterion.

Radiology AI publication counts

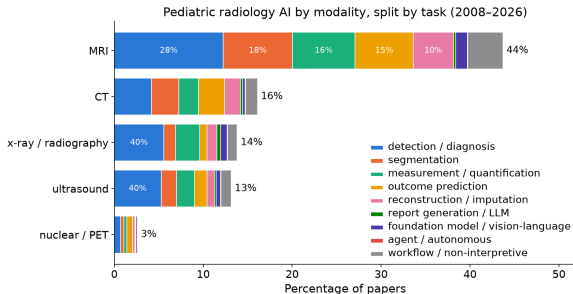


Which journals publish pediatric radiology AI



x = OpenAlex 2-year mean citedness (a free JIF analogue; runs below the published JIF). y = cumulative pediatric radiology-AI papers 2015-2026 (PubMed, strict title/abstract query). The 29 journals with at least 25 of them are shown, carrying 1,511 of 4,399; the rest is a long tail. Largest: Scientific Reports (156); median impact 3.8, 1 at 10 or above – this work lands mostly in mid-impact specialty and technical journals. A further 3 conference proceedings and preprint servers clear the threshold (95 papers) but have no impact number of any kind and are not plotted. The same chart animated year by year: [figures/journal_impact.gif](#).

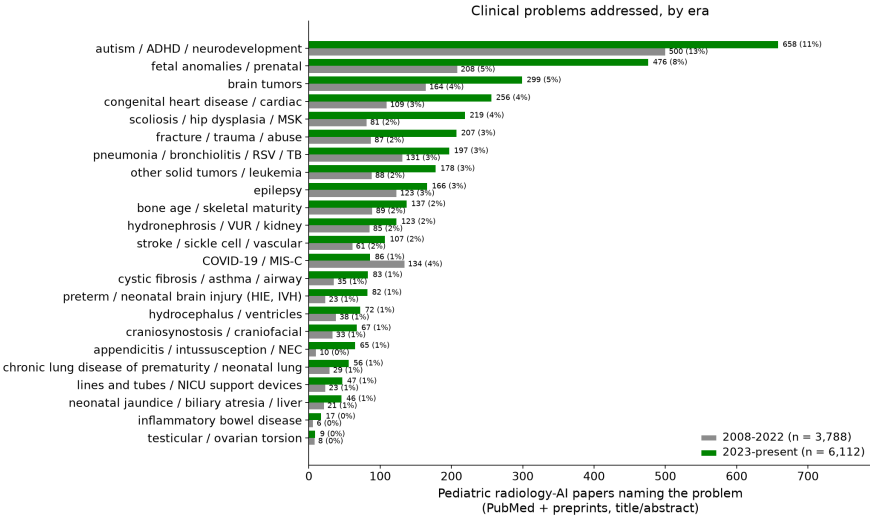
Pediatric radiology AI publications, stratified by modality and task



Same construction, restricted to the pediatric subset (9,762 papers incl. 608 preprints). Top pediatric modalities: MRI 41%, CT 16%, x-ray / radiography 13%, ultrasound 12%. Mammography is omitted: its 67 pediatric-query hits (of 9,762) are adult breast-imaging papers whose abstracts mention children.

- detection / diagnosis:** is a finding present now, and which one (label or box): fracture, pneumonia, tumor; includes triage and screening
- segmentation:** outline a structure or lesion (a mask)
- measurement / quantification:** a number from the image: bone age, Cobb angle, organ volume, fetal biometry
- outcome prediction:** a future risk, response, or survival estimate from the image (prognosis, radiomics signatures)
- reconstruction / imputation:** a better or missing image: lower dose, faster scans, denoising, synthetic CT/MR
- report generation / LLM:** text: draft, summarize, or extract from radiology reports
- foundation model / vision-language:** a large pretrained model reused across tasks (segment-anything, self-supervised)
- agent / autonomous:** multi-step actions taken without a human in the loop
- workflow / non-interpretive:** protocoling, scheduling, ordering, decision support, education

Pediatric radiology AI publications, stratified by medical problem



Most-cited radiology AI papers, 2008-2022

Cites	FWCI	Paper	Venue	What it answers
5,797	443.4	Swin-Unet: Unet-like Pure Transformer for Medical Image Segmentation (2021)	ECCV Workshops	<i>segmentation</i>
2,628	288.8	Swin UNETR: Swin Transformers for Semantic Segmentation of Brain Tumors in MRI Images (2022)	BrainLes@MICCAI	<i>segmentation</i>
2,782	262.1	COVID-Net: a tailored deep convolutional neural network design for detection of COVID-19 cases from chest X-ray images (2020)	Scientific Reports	<i>classification</i> : Can a CNN detect COVID-19 on chest X-ray? (COVID-Net)
3,710	230.9	ChestX-Ray8: Hospital-Scale Chest X-Ray Database and Benchmarks on Weakly-Supervised Classification and Localization of Common Thorax Diseases (2017)	Computer Vision and Pattern Recognition	<i>classification</i>
10,389	219.2	nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation (2020)	Nature Methods	<i>segmentation</i> : Can one self-configuring method segment any organ or lesion without hand tuning? (nnU-Net)
2,036	206.6	Covid-19: automatic detection from X-ray images utilizing transfer learning with convolutional neural networks (2020)	Physical and Engineering Sciences in Medicine	<i>classification</i> : Can transfer learning detect COVID-19 on X-ray with small data?
5,662	156.0	Computational Radiomics System to Decode the Radiographic Phenotype (2017)	Cancer Research	<i>radiomics</i> : Can standardized image features (radiomics) be extracted reproducibly? (PyRadiomics)
2,145	135.7	MultiResUNet : Rethinking the U-Net Architecture for Multimodal Biomedical Image Segmentation (2019)	Neural Networks	<i>segmentation</i> : Architecture variant of U-Net for multimodal segmentation
2,126	119.7	CE-Net: Context Encoder Network for 2D Medical Image Segmentation (2019)	IEEE Transactions on Medical Imaging	<i>segmentation</i> : Architecture variant for 2D medical image segmentation (CE-Net)
9,731	86.7	UNet++: A Nested U-Net Architecture for Medical Image Segmentation (2018)	DLMIA/ML-CDS@MICCAI	<i>segmentation</i>

FWCI = field-weighted citation impact (OpenAlex): citations relative to the world average for papers of the same field and year (1 = average)

Most-cited radiology AI papers, 2026 (year to date)

Cites	FWCI	Paper	Venue	What it answers
59	111.4	A generalizable foundation model for analysis of human brain MRI	Nature Neuroscience	<i>other</i>
12	61.4	Explainability-Aligned Reliability-Weighted Fuzzy Ensemble for Automated Cervical Cancer Classification	International Journal of Intelligent Systems	<i>oncology</i>
21	58.8	A hybrid deep learning framework based on VGG19 and U-Net for accurate brain tumor segmentation in MRI images	Magnetic Resonance Letters	<i>segmentation</i>
14	50.1	MedVersa: A Generalist Foundation Model for Diverse Medical Imaging Tasks	NEJM AI	<i>other</i>
15	49.5	Explainable AI-driven MRI-based brain tumor classification: a novel deep learning approach	Frontiers Artif. Intell.	<i>oncology</i>
13	46.0	A deep learning framework for breast cancer diagnosis using Swin Transformer and Dual-Attention Multi-scale Fusion Network	Scientific Reports	<i>oncology</i>
13	44.2	Multi-class eye disease classification using deep learning EfficientNetB0 fusion techniques	Scientific Reports	<i>classification</i>
12	39.6	3D-UNet+: An enhanced 3D U-Net framework for brain tumor segmentation with systematic hyperparameter tuning	Neural computing & applications (Print)	<i>segmentation</i>
13	30.9	Deep Learning Approaches for Brain Tumor Classification in MRI Scans: An Analysis of Model Interpretability	Applied Sciences	<i>oncology</i>
19	27.1	BIBSNet: A deep learning baby image brain segmentation network for MRI scans	Developmental Cognitive Neuroscience	<i>segmentation</i> : Can one network segment infant brain MRI across the first years of life? (BIBSNet)

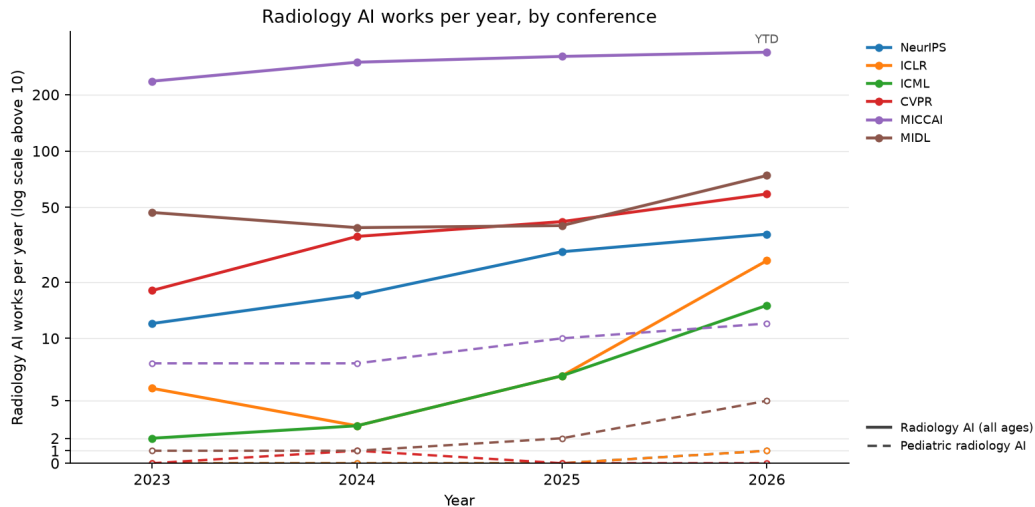
Most-cited pediatric radiology AI papers, 2008-2022

Cites	FWCI	Paper	Venue	What it answers
426	310.7	Performance of a Deep-Learning Neural Network Model in Assessing Skeletal Maturity on Pediatric Hand Radiographs. (2017)	Radiology	<i>bone age</i> : Can a CNN assess skeletal maturity as accurately as radiologists?
432	305.5	Fully Automated Deep Learning System for Bone Age Assessment (2017)	Journal of digital imaging	<i>bone age</i> : Fully automated bone age from hand radiographs
407	206.8	Deep learning for automated skeletal bone age assessment in X-ray images (2017)	Medical Image Anal.	<i>bone age</i> : Deep learning for automated bone age (BoNet)
144	69.0	Computerized Bone Age Estimation Using Deep Learning Based Program: Evaluation of the Accuracy and Efficiency. (2017)	AJR. American journal of roentgenology	<i>bone age</i> : Is a commercial deep-learning bone-age program accurate and faster in practice?
353	55.0	Standard Plane Localization in Fetal Ultrasound via Domain Transferred Deep Neural Networks (2015)	IEEE journal of biomedical and health informatics	<i>fetal MRI</i>
196	54.4	Automatic Fetal Ultrasound Standard Plane Recognition Based on Deep Learning and IoT (2021)	IEEE Transactions on Industrial Informatics	<i>fetal MRI</i>
449	35.4	A transfer learning method with deep residual network for pediatric pneumonia diagnosis (2020)	Comput. Methods Programs Biomed.	<i>pneumonia / COVID</i>
236	28.9	Evaluation of deep convolutional neural networks for automatic classification of common maternal fetal ultrasound planes (2020)	Scientific Reports	<i>fetal MRI</i>
460	26.4	The RSNA Pediatric Bone Age Machine Learning Challenge. (2019)	Radiology	<i>bone age</i> : How well can crowdsourced AI estimate bone age? (RSNA challenge, 260 teams)
258	25.8	An ensemble of neural networks provides expert-level prenatal detection of complex congenital heart disease (2021)	Nature Medicine	<i>detection</i>

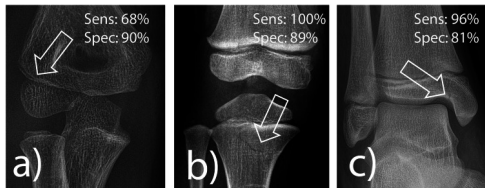
Most-cited pediatric radiology AI papers, 2026 (year to date)

Cites	FWCI	Paper	Venue	What it answers
3	45.1	Wavelet-Based Frequency Replacement and Edge Enhancement for Semi-Supervised Fetal Ultrasound Image Segmentation	Journal of ultrasound in medicine	<i>segmentation</i>
19	27.1	BIBSNet: A deep learning baby image brain segmentation network for MRI scans	Developmental Cognitive Neuroscience	<i>segmentation</i> : Can one network segment infant brain MRI across the first years of life? (BIBSNet)
4	27.1	Beyond Benchmarks of IUGC: Rethinking Requirements of Deep Learning Methods for Intrapartum Ultrasound Biometry from Fetal Ultrasound Videos	Medical Image Anal.	<i>fetal MRI</i>
3	26.2	Zero-shot performance of a general-purpose vision-language model for pediatric appendicitis diagnosis.	Pediatric Radiology	<i>classification</i>
5	24.1	Deep learning assessment of fetal brain maturation on 3D ultrasound volumes in early-onset fetal growth restriction	Ultrasound in Obstetrics and Gynecology	<i>fetal ultrasound</i> : Can deep learning grade fetal brain maturation on 3D ultrasound in growth restriction?
3	18.1	A multi-task deep learning and radiomics framework for fetal anatomical structure detection and classification in ultrasound imaging	Scientific Reports	<i>radiomics</i>
2	18.1	Deep learning-based cerebellar segmentation on T2-weighted fluid-attenuated inversion recovery magnetic resonance imaging for detecting cerebellar hypoplasia/atrophy in infants	Pediatric Radiology	<i>segmentation</i>
3	14.6	Deep Learning-Based Multi-Class Pediatric Wrist Fracture Subtype Classification: A Pilot Study Comparing Convolutional Neural Network Architectures	Journal of Imaging	<i>fracture</i>
4	9.8	Deep learning super-resolution of paediatric ultra-low-field MRI without paired high-field scans	bioRxiv	<i>other</i>
2	9.8	Acquisition time/dose reduction in pediatric PET imaging using patch-based deep learning	EJNMMI Physics	<i>reconstruction</i> : Can pediatric PET use one fifth of the counts with patch-based deep-learning denoising?

Radiology AI works per year, by conference



AI fracture detection in a real pediatric emergency department

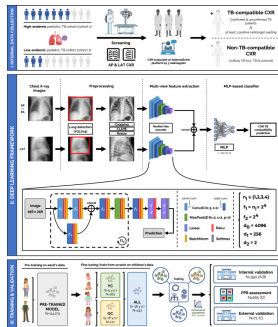


Ziegner et al., Eur Radiol 2025, figure 4: pediatric fractures the software found, with its per-region sensitivity and specificity

source: cdn.ncbi.nlm.nih.gov

- **Problem:** fractures in children are missed most often by the least experienced readers, at night, in the emergency department.
- **Data:** 1,672 consecutive radiographs of children under 18 from one tertiary pediatric emergency department, plus a selected medicolegal set (proximal tibia, radial condyle); three pediatric residents read each before and after seeing the AI.
- **Method:** a commercially available, CE-marked deep-learning fracture detector (adult-derived, pediatric extension) run stand-alone and as a second reader.
- **Result:** stand-alone sensitivity 92%, specificity 83%; 100% for proximal tibia but 68% for radial condyle fractures; residents' accuracy rose from 88% to 90%, and in 2% of cases they dropped a correct diagnosis after seeing the AI.
- **Why it matters:** the first real-life pediatric ED evaluation: the gain from a cleared adult tool is real but modest, and the elbow remains the weak spot.

pTBLightNet: pediatric tuberculosis on frontal and lateral chest radiographs



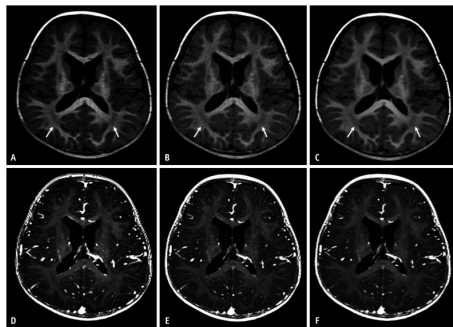
Capellan-Martin et al., Nat Commun 2025, figure 1: the multi-view

pTBLightNet pipeline

source: cdn.ncbi.nlm.nih.gov

- **Problem:** pediatric pulmonary TB has vaguer symptoms and subtler radiographs than adult TB; most affected children in high-burden settings are never diagnosed.
- **Data:** pre-trained on 114,173 adult chest radiographs, then fine-tuned and tested on 918 radiographs from three pediatric TB cohorts, with lateral views.
- **Method:** a multi-view network that scores frontal and lateral films for TB-compatible findings; separate models for under-5 and 5-18 years.
- **Result:** AUC 0.90 on internal and 0.68 on external testing; lateral views helped most in the youngest children; age-specific models matched larger undifferentiated ones.
- **Why it matters:** shows both the value of adult pre-training for a rare pediatric target and the internal-to-external drop that every pediatric deployment should expect.

Deep-learning reconstruction shortens pediatric brain MRI



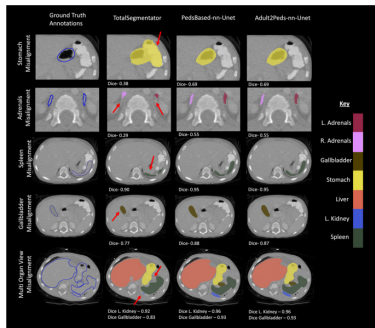
Yoo et al., Korean J Radiol 2025, figure 2: conventional (A, D) vs accelerated 3D T1 pediatric brain MRI without (B, E) and with (C, F)

deep-learning reconstruction

source: cdn.ncbi.nlm.nih.gov

- **Problem:** long 3D T1 sequences are where children move, sedation is prescribed, and scanner time is lost.
- **Data:** 46 children scanned with both the conventional and an accelerated 3D T1 protocol (pre- and post-contrast) on one 3T scanner.
- **Method:** vendor deep-learning reconstruction applied to the accelerated acquisition; noise, contrast, signal-to-noise and radiologist ratings compared with the conventional scan.
- **Result:** acquisition time cut by 29% (pre-contrast) and 41% (post-contrast); the deep-learning images had lower noise, higher SNR and CNR, better overall quality and fewer artifacts than the conventional scan.
- **Why it matters:** reconstruction AI is on the scanner today and gives children a shorter, better exam without any diagnostic model in the loop.

Children are not small adults: an adult CT segmentation model in children



Chatterjee et al., J Imaging Inform Med 2025, figure 7: pediatric CT

organ labels, ground truth vs adult-trained TotalSegmentator vs

pediatric-trained models

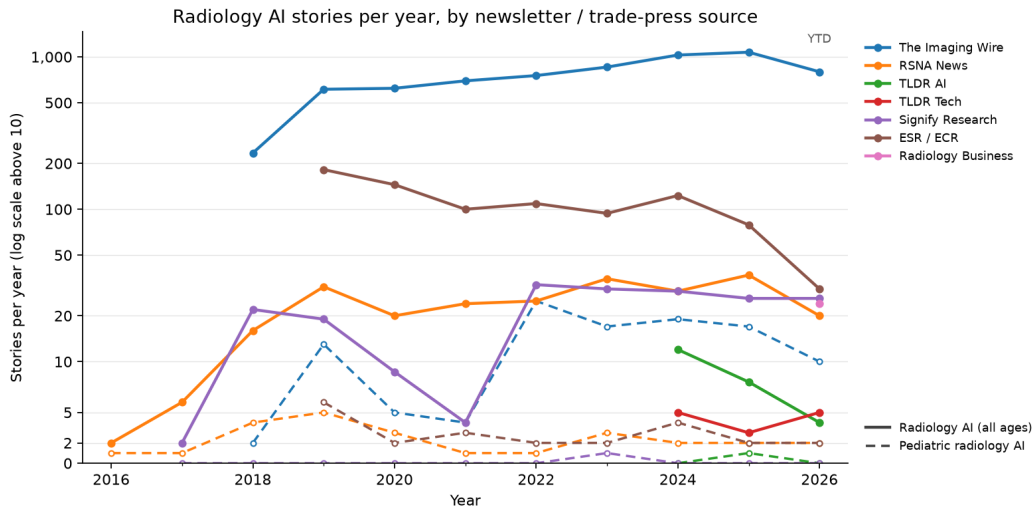
source: cdn.ncbi.nlm.nih.gov

- **Problem:** the most-used open CT organ segmentation model (TotalSegmentator) was trained on adults; nobody had measured how it behaves in children.
- **Data:** external adult ($n = 300$) and pediatric ($n = 359$) abdominal CT sets with expert organ labels.
- **Method:** TotalSegmentator scored on both sets; then a pediatric-only nnU-Net and an adult model fine-tuned on pediatric cases.
- **Result:** mean Dice fell from 0.81 in adults to 0.73 in children, worst for the adrenals (0.69 to 0.41 and 0.35), duodenum and pancreas; pediatric training or fine-tuning recovered the loss.
- **Why it matters:** the clearest quantitative case for local pediatric validation and fine-tuning of adult-derived tools before clinical use.

The biggest public pediatric radiology AI datasets

Dataset	Year	Modality	Size	Ages	Access
CBTN clinical MRI (Children's Brain Tumor Network)	2023	brain / spine tumor MRI	23,101 exams (1,526 patients); tumor masks for 370	pediatric	application (NCI CCDI)
GRAZPEDWRI-DX	2022	wrist radiographs (trauma)	20,327 images (6,091 patients)	0.2–19 y	open
RSNA Pediatric Bone Age Challenge	2017	hand radiographs	14,236 images (bone age + sex)	mean 10.6 y	open (RSNA terms)
FETAL_PLANES_DB	2020	fetal ultrasound (standard planes)	12,400 images (1,792 pregnancies)	fetal	open
ABCD Study	2018–	brain MRI, longitudinal	11,878 participants, up to 7 follow-ups	9–10 y at baseline	application (NBDC)
PediCXR / VinDr-PCXR	2023	chest radiographs	9,125 studies; 36 findings, 15 diagnoses	< 10 y	credentialed (PhysioNet)
Guangzhou pediatric chest X-ray (Kermany)	2018	chest radiographs	5,856 images (normal / bacterial / viral pneumonia)	1–5 y	open
ABIDE I + II (autism)	2012 / 2016	brain MRI (structural, resting fMRI)	2,156 subjects	5–64 y (mostly children)	open
Regensburg Pediatric Appendicitis	2023	abdominal ultrasound + labs	1,709 images (579 patients)	0–18 y	open
dHCP (Developing Human Connectome Project)	2022–24	neonatal + fetal brain MRI	886 neonatal + 297 fetal datasets, with segmentations	20–45 weeks	open (NDA)
Pediatric-CT-SEG (TCIA)	2022	chest / abdomen / pelvis CT	718 series (359 patients); 29 organ contours	5 days – 16 y	open
BraTS-PEDs (pediatric brain tumor challenge)	2023–25	brain tumor MRI	464 patients (2024 challenge); 1,649 series on TCIA	pediatric	registration / TCIA
FeTA (fetal brain tissue annotation)	2021–24	fetal brain MRI	280 volumes (120 train + 160 test); 7 tissue labels	18–35 weeks	registration

Radiology AI newsletter stories per year, by source



Pediatric radiology AI in the trade press, 2026 (year to date) (14 stories)

Date	Newsletter	Story	Paper
2026-01-07	The Imaging Wire	BrightHeart Raises EUR 11M	
2026-01-28	The Imaging Wire	AHA Publishes New Stroke Guidelines	Prabhakaran et al., 2026 (Stroke)
2026-01-28	ESR / ECR	Guerbet: Innovation and French Cooperation in the Spotlight at ECR 2026	
2026-02-01	The Imaging Wire	DL-Based CT Reduces Radiation Dose	
2026-02-08	The Imaging Wire	GE Bolsters Ultrasound with Fetal AI	
2026-02-26	ESR / ECR	MRI Unveiling at the European Congress of Radiology (ECR) 2026	
2026-03-29	The Imaging Wire	Ultrasound AI Gets Breakthrough Nod	
2026-05-10	The Imaging Wire	Pediatric MRI Safety, RTs and Remote Radiology, and PSMA-PET	
2026-06-22	RSNA News	Researchers Use AI to Establish Body Charts for CT Imaging Across the Adult Lifespan	
2026-06-28	The Imaging Wire	BrightHeart's Bright Future	Zelop et al., 2025 (Obstetrics & Gynecology)
2026-07-12	The Imaging Wire	AI's Rise in Pediatric Imaging	Shah et al., 2026 (European Radiology)
2026-08-16	The Imaging Wire	CT vs. MRI for Pediatric TBI, Helium Vulnerability, and Ditching the Disk	
2026-08-25	RSNA News	AI Challenges Fuel Innovation Beyond the Leaderboard	
2026-09-23	The Imaging Wire	Repeat MRI Scans Take Operational Toll	

Story 3: what pediatric departments actually report using

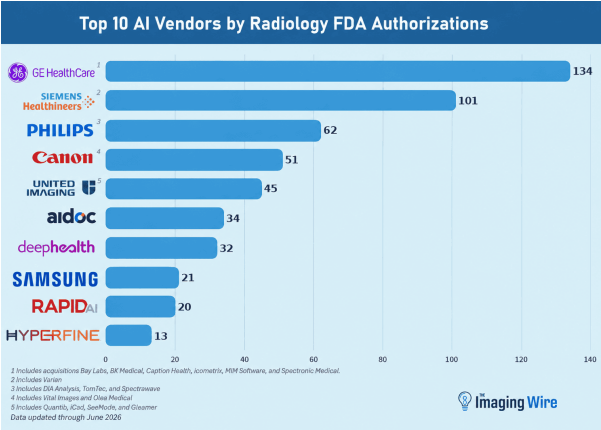
The Imaging Wire, 12 July 2026: “AI’s Rise in Pediatric Imaging” — Shah et al., *European Radiology* 2026 (doi 10.1007/s00330-026-12728-9)

- **Why it was news:** after a decade of radiology AI research, almost no data existed on what pediatric departments have actually deployed. This international survey of 18 academic pediatric radiology centers (69% response) is the first real answer.
- **Adoption is wide but shallow:** **88.9%** run at least one AI tool, yet the tools cluster on two tasks — bone age (44.4%) and segmentation / quantification (22.2%) — and only **16.7%** called AI “transformational”.
- **The barriers named:** lack of pediatric datasets (83.3%), integration into PACS and workflow (66.7%), cost (50%).
- **It matches this deck’s own data:** bone age is also the most common pediatric clinical problem in the published literature and the one task with a mature commercial product.

Take-home: the constraint is not model accuracy. It is pediatric training data and integration — both of which a children’s hospital, not a vendor, is positioned to fix.

Radiology keeps its lead in FDA AI authorizations

The Imaging Wire, 14 September 2026: FDA AI-enabled device list, updated through June 2026



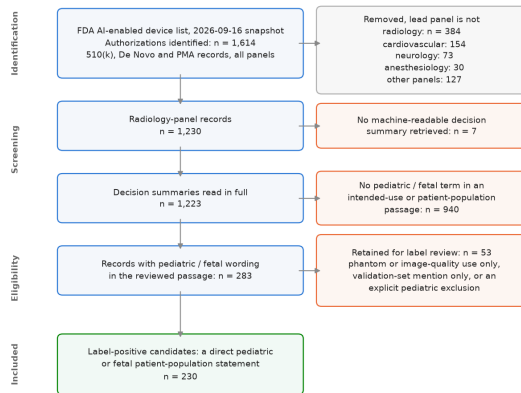
- **1,614** AI-enabled devices authorized since the FDA began tracking, up 5.9% from Q1 2026.
- **1,230** of them (76%) are radiology, the same radiology-panel total this deck's own FDA pull finds.
- Q2 2026: 89 authorizations (92 in Q1), 66 of them (74%) radiology.
- Scanner makers lead: GE HealthCare 134, Siemens Healthineers 101, Philips 62, Canon 51, United Imaging 45.
- The largest software-only vendors are Aidoc (34), DeepHealth (32) and RapidAI (20).
- The FDA is still deciding how to regulate generative AI used in direct clinical care.

Top 10 vendors by radiology FDA authorizations, counts include acquisitions

source: theimagingwire.com

How many cleared devices actually name children?

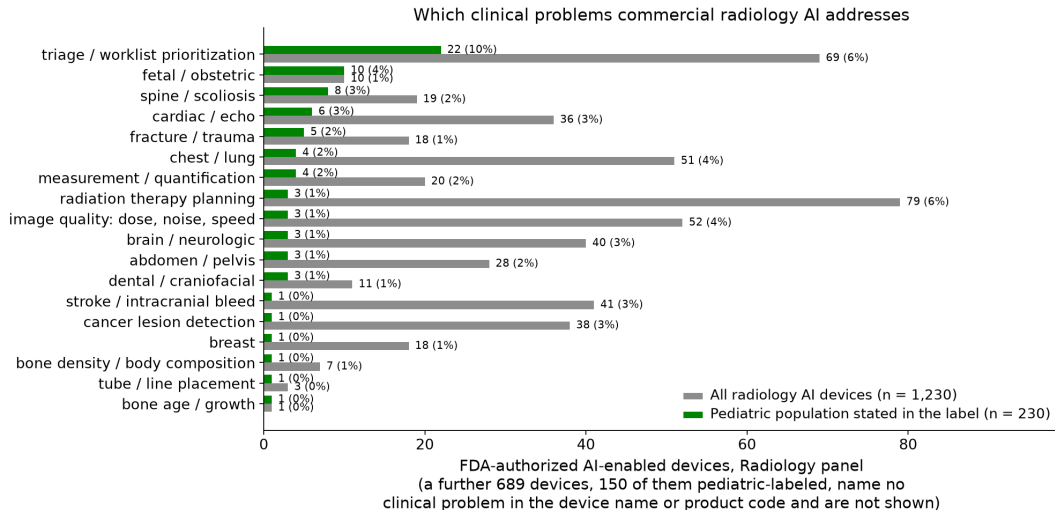
FDA AI-enabled device list: which authorizations name children



Keyword screen of the linked FDA decision summaries, not a determination of pediatric authorization:
confirm the current labeling before purchase. One row per submission, so versions and resubmissions count separately.

230 of the 1,230 radiology-panel authorizations (19%) name a pediatric or fetal population in the labeling passage reviewed; 53 more name children only as a phantom, a validation set or an exclusion, and are kept for label review.

Which clinical problems the cleared products address



Product 1: BoneXpert — the most-validated pediatric AI you cannot buy

Visiana, Denmark (bonexpert.com) — bone age / growth

- **What it does:** automated Greulich–Pyle and Tanner–Whitehouse bone age from a single hand radiograph, plus a **Bone Health Index** from cortical thickness and an adult-height predictor. Not a second reader — it replaces the manual rating.
- **Why it is the benchmark:** designed for children from the start, in routine European clinical use since 2009, and the comparator nearly every academic bone-age paper measures itself against. It is the one pediatric AI with a decade of unglamorous production experience behind it.
- **The catch:** CE-marked in Europe; in the United States the vendor labels it **research use only**. There is no FDA authorization. The only pediatric-specific US bone-age clearance is a different product, EFAI Bonesuite XR Bone Age Pro (ages 2–16, K234042, 2024).
- **What it tells us about the market:** pediatric-native tools exist and work. The bottleneck is a US regulatory and commercial path, not the science.

Take-home: when someone says “there is no pediatric AI”, the accurate statement is that the best pediatric AI is not licensed here.

Product 2: BoneView — read the anatomy, not just the age range

Gleamer, France — fracture / trauma; US clearance K222176 (BoneView 1.1-US, 2023)

- **What it does:** flags suspected fractures on trauma radiographs as regions of interest beside the image. The international suite also covers dislocations, effusions and focal bone lesions; **the US clearance is fracture detection only.**
- **The indication is narrower than “ages 2–21”:** K222176 permits ages 2–21 for **specified extremity views**, clavicle included. Pelvis, hip, femur, ribs and spine are **adult-only** in that summary — a child with a suspected hip or rib fracture is outside the pediatric indication.
- **What the real-life pediatric evidence shows:** the one pediatric ED evaluation of a cleared fracture detector (Ziegner et al., *Eur Radiol* 2025, 1,672 radiographs) found 92% stand-alone sensitivity but only **68% for radial condyle fractures**, and residents' accuracy rose 88% → 90%.
- **The failure mode to watch:** in 2% of cases readers abandoned a correct diagnosis after seeing the AI.

Take-home: the most mature pediatric AI market still needs local validation by anatomy, and a reader policy for disagreement.

Product 3: Aidoc BriefCase — a pediatric label is not pediatric evidence

Aidoc, Israel — CT/CTA triage and worklist prioritization; e.g. K253578 (2026)

- **What it does:** reads CT and CTA as they arrive and reorders the worklist — intracranial hemorrhage, pulmonary embolism, pneumothorax, cervical-spine and rib fractures, free gas. Triage, not diagnosis, is the product.
- **The number that matters here:** Aidoc holds **23 of the 230 pediatric-label-positive** radiology authorizations in our FDA screen, more than any other company.
- **What that does and does not mean:** the decision summaries state a pediatric patient population, but the **performance data behind them is adult**. A population statement in an intended-use paragraph is a candidate for label review, not evidence of testing in children.
- **Why it matters operationally:** triage silently changes the order a pediatric radiologist sees cases in. A model tuned on adult prevalence will mis-prioritize a pediatric list, and nobody sees the ranking it produced.

Take-home: “FDA-cleared, pediatric population included” is this market’s most common and most misleading claim. Ask for pediatric subgroup performance; usually there is none.

Product 4: deep-learning reconstruction — the AI children actually meet

GE AIR Recon DL / TrueFidelity; Siemens Deep Resolve; Canon AiCE; Philips SmartSpeed / Precise Image

- **What it does:** reconstructs MRI and CT from noisier, faster or lower-dose raw data — shorter MRI at comparable quality, CT at reduced dose. It is not interpretive: it makes the picture the radiologist then reads.
- **Why it is the biggest pediatric win on this list:** a shorter MRI is **fewer sedations**; a lower-dose CT is **less radiation to a growing child**. Neither requires anyone to trust a diagnostic model.
- **The regulatory shape is different:** these are sold across ages as part of the scanner, with vendor-published pediatric protocols rather than a pediatric indication. Philips holds 28 pediatric-label-positive entries, **one** of which is software; the rest are scanner labels listing pediatric imaging as an application.
- **The caveat:** dose and time reductions are not fixed properties of the product name — they depend on task, patient size and configuration, and must be measured locally.

Take-home: most of the pediatric benefit already installed here is non-interpretive AI nobody calls AI, and it is where a validation program should start.

Kamran review: broader research, stronger validation

Kamran et al., *Pediatric Radiology* 2026 (doi 10.1007/s00247-025-06462-5)

- **A large but concentrated literature:** 789 pediatric-focused original articles (2005–August 2024); 91% addressed image interpretation/diagnosis. Leading applications were bone age (88), pneumonia (65), and brain lesions/tumors (54).
- **Generalizability needs stronger evidence:** 65% relied on local hospital datasets. The authors call for diverse, multicenter datasets and prospective validation across pediatric ages, sites, and disease-prevalence settings.
- **Broaden the research agenda:** communication, education, policy, and stakeholder perspectives were underrepresented. Include children, families, and clinical teams in development and evaluation.

Take-home: prioritize representative data, validation, and neglected clinical needs. This review maps research activity; it does not compare model accuracy or establish clinical readiness.

ACR white paper: the pediatric AI gap

Sammer et al., *J Am Coll Radiol* 2023;20:730–737 (doi 10.1016/j.jacr.2023.06.003)

- **Scope:** ACR Pediatric AI Workgroup white paper framing limited access to pediatric AI as a health equity issue.
- **Availability:** at the time of the review, only 6 of more than 200 FDA-cleared tools in the ACR catalog indicated pediatric use (about 3%).
- **Why adult AI may fail:** growth, disease patterns, and imaging protocols differ; published studies show uneven performance across pediatric subgroups.
- **Barriers:** scarce labeled pediatric data, consent and privacy concerns, and weak financial incentives for smaller pediatric markets.
- **Recommendations:** validate in children, disclose age-specific evidence, fund pediatric development, involve families, and protect pediatric workflows. The authors introduce *Image IntelliGently* to advance this work.

Multisociety statement: safe pediatric AI implementation

Shelmerdine et al., *J Am Coll Radiol* 2026;23:89–101 (doi 10.1016/j.jacr.2025.08.019)

ACR, ESPR, SPR, SLARP, AOSPR, and SPIN propose four pillars for safe adoption:

- **Regulation and purchasing:** pediatric labeling and proposed safety ratings; diverse datasets, subgroup performance, transparency, and cybersecurity.
- **Implementation and integration:** define a local clinical need, select tools validated in children, pilot before scaling, and involve pediatric specialists and families.
- **Interpretation and postmarket surveillance:** retain human oversight; audit performance and patient outcomes; manage discrepancies, report adverse events, and address automation bias.
- **Education:** foundational AI literacy for all staff, pediatric-specific training, continuing education, and patient/public engagement.

Worth knowing as a radiologist

- **Deep Learning: A Primer for Radiologists** (Radiographics paper, 2017): a helpful introduction to the principles and applications of deep learning in radiology; essential for understanding how AI tools work and their limitations
- **RSNA Pediatric Bone Age Challenge (2017)** (benchmark / dataset): a landmark pediatric benchmark; useful for understanding automated bone-age accuracy and its limits
- **Gleamer BoneView / AZmed Rayvolve** (clinical products): examples of fracture-detection assistance with pediatric indications; eligibility depends on age, anatomy, and product version
- **FDA AI-enabled device list and linked labeling** (regulatory resource): a starting point for checking authorized use and pediatric age ranges; the list is not exhaustive
- **CLAIM checklist (2024 update)** (reporting guideline): check whether data, reference standards, and testing are adequately described; reporting completeness alone does not establish clinical benefit
- **TotalSegmentator** (research tool): automated CT / MR organ segmentation for volumes and body composition; check accuracy in children before trusting measurements
- **nnU-Net** (research framework): a strong baseline for training segmentation models; ask whether a new method improves on an appropriate comparator
- **MedSAM / segment-anything models** (research models): prompt-guided contours can assist annotation; radiologists still need to review and correct the output
- **MONAI** (research framework): infrastructure for imaging-AI development; useful when collaborating with a research team, rather than a ready-to-use diagnostic tool
- **NVIDIA NV-Reason-CT** (open research model, Sept 2026): a 3D CT vision-language model that drafts chest / abdominal CT findings with step-by-step reasoning and answers questions; open weights, research use only, not FDA-cleared, and no pediatric data or age range reported

Placeholder: database screenshot
(replace figures/explore_database.png)

Recommendations for pediatric radiologists

- AI development is constrained by scarce public datasets, fragmented institutional data, rare diseases, and heterogeneous reporting practices.
- Choose one task worth improving in your practice: bone-age measurements, fracture detection, or reducing CT dose / MRI scan time.
- Before adopting a tool, ask which children it was tested on. Check the intended age range and test it on your own cases, including younger children and unusual anatomy.
- In a local pilot, measure what changes with AI: reading time, missed findings, false alarms, dose, or scan time. Review errors and keep checking after rollout.
- Help build the evidence we need: choose a pediatric problem you see often, label cases carefully, and work with other hospitals to test whether a model works beyond the site that built it.